SUPPORTING ADULT GUIDANCE

Reverse percentages

Percentages, Ratio, Growth and Bounds · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with a sale tag (20% off, now £48): learners test guesses by running them forwards. The main activity, the Undo Machine, is in Reflection. Learners choose a situation (a sale price, a price rise or a plant's height), set the percentage change and the value after it, and may type a prediction before pressing Run backwards. The machine then shows the original as the after value over the multiplier (a stacked fraction), checks it forwards, and draws bars for the original and the after value. Show the slip draws a third bar: the percentage taken of the after value, which never checks forwards. Every value on screen is exact; where an original is not a whole number of pence, the machine says so. At –100% it explains why the change cannot be undone. Six questions, a stretch (why 20% off £96 misses), true-or-not statements, quick undo and puzzle briefs follow.

Driving Question: *How do I undo a change?*

The mathematics, in plain terms

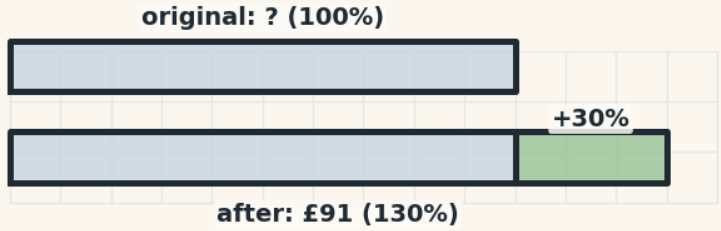
If a change of p% gives an after value A, then $A = O(1 + p/100)$, so $O = A/(1 + p/100)$. The slip computes $A(1 – p/100)$ instead, and $A(1 – p/100)(1 + p/100) = A(1 – p^2/10000)$, which is less than A for any non-zero p: the slip always fails the forwards check. Dividing by m is the same as multiplying by 1/m, so the percentage that undoes a change is not the same number: 1/1.25 = 0.8, so +25% is undone by –20%.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Undoing an increase

After a 30% rise, a bike costs £91. What was the original price?

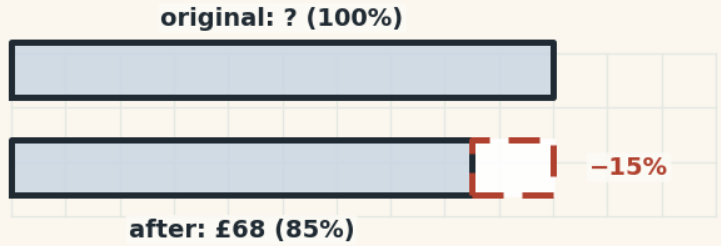


1. Which value do you have?	£91 is the value after the rise
2. The multiplier	100% + 30% = 130% = 1.3
3. Undo by dividing	$\frac{91}{1.3} = 70$
4. Check forwards	£70 × 1.3 = £91

The original price was £70 (not £91 – 30% of £91 = £63.70).

Example 2 — Undoing a decrease

After 15% off in a sale, a coat costs £68. What was the original price?



1. The multiplier	100% – 15% = 85% = 0.85
2. Undo by dividing	$\frac{68}{0.85} = 80$
3. Check forwards	£80 × 0.85 = £68

The original price was £80 (not £68 + 15% of £68 = £78.20).

Predict, run backwards, then check forwards

Learners first test guesses forwards, so that checking forwards is a habit before any method appears. The machine then reveals the original only when asked, after an optional prediction. Showing the slip beside the correct original, and running it forwards, lets learners see the error for themselves rather than being told.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	What was it before? Test guesses for a sale price forwards; the idea card.	~5 min
Movement	Quick undo: eight originals that work in your head.	~6 min
Reflection	The Undo Machine (main activity), six questions, the stretch, and true or not.	~20 min
Creativity	Undo puzzles: six briefs.	~9 min
Rest	Undoing is not always the same step backwards.	~2 min
Nutrition	Practical: checking sale prices, VAT and bills.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Taking the percentage of the after value (the slip): £96 – 20% of £96.
Ask: “Twenty per cent of which amount?”

Multiplying by the multiplier again instead of dividing.
Ask: “Is the value you have before or after the change?”

Using the wrong multiplier for a decrease (1.15 instead of 0.85).
Ask: “What percentage of the original is left?”

Expecting the same percentage to undo the change.
Ask: “Try it with £100. Does it come back?”

Not checking the answer.
Ask: “If that were the original, what would the change give?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: percentages as multipliers (Lesson 3) and percentage change (Lesson 4). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a coat costing £84 after a 20% rise must have been £67.20. The aim is a question that sends them forwards to check.

- “If the coat was £67.20, what would a 20% rise make it?”

- “Is that £84?”
- “So the 20% was a percentage of which amount?”
- “What do you multiply the original by to get £84? So how do you undo that?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the machine sometimes show a fraction such as £43 11/23?

So that every value on screen is exact. It also says when no real price (a whole number of pence) could give that after value. Learners' rounded answers are accepted.

Can learners use the unitary method instead?

Yes. Finding 1% (or 10%) and building back to 100% is the same idea in steps; the machine's readout shows it too. Dividing by the multiplier does it in one step.

Is the stretch required?

No. The secure core is dividing the after value by the multiplier and checking forwards.

What comes next?

Lesson 6: compound interest and depreciation, repeating the same multiplier.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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